

Solution Requirements

Team Leader: Gayathri A **Members:** Srinivasan E, Saravanavel V

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	No authentication in current scope; future version can add student accounts.	Low
2	Authorization	No role-based access in current stateless version.	Low
3	External Interfaces	Integrate with Google Gemini through the Google GenAI SDK.	High
4	Transactions Processing	Accept academic input, generate AI response, parse/return structured result.	High
5	Reporting	Provide readable outputs and API documentation through Swagger UI; persistent reports are future scope.	Medium
6	Business Rules	Feature-specific prompts; quiz question count configurable; model fallback when primary model is unavailable.	High
7	Compliance / Privacy	Keep API credentials in environment variables; avoid exposing secret keys in frontend/source code.	High
8	Other	Offline/demo mode, five academic workflows, responsive web interface, REST endpoints.	High

NFR Category	Requirement Description	Acceptance Criteria
Performance & Speed	Return results without unnecessary processing overhead.	Normal requests should complete within practical API/network limits.
Scalability	Modular feature/service architecture should permit additional users/features.	Features remain independently maintainable and deployable.
Security & Data Privacy	Secrets stored through environment configuration.	No Gemini API key hard-coded in client-side code.
Reliability & Availability	Fallback to alternate Gemini model or offline/demo mode.	Application should not crash solely because a configured model is unavailable.
Usability & Accessibility	Simple web interface with clear inputs and outputs.	A student can select a feature and submit a request without constructing complex prompts.
Maintainability / Portability	Python/FastAPI modular code with documented dependencies.	Runs on Windows/macOS/Linux with Python 3.10+.